

Also, select a reference resistance that maximizes the usable input range of the ADC. As an example, it helps to show this with several values. Start with a 2-wire RTD ratiometric measurement with a PT100 where the maximum resistance is 400 Ω . This is the setup shown for a basic ratiometric measurement in [Figure 1-4](#).

If the IDAC current is selected to be 1 mA, then the reference resistor could be chosen to be 1620 Ω . The measurement of the 400 Ω could be set to a PGA gain of 4. This would make the input voltage 1.6 V, while the reference voltage is set to 1.62 V. This would maximize the input voltage range of the ADC to 98.8% of the positive full-scale range. A reference resistor of 1600 Ω could have been chosen to maximize the ADC, however a small gain error or resistance error may push a 400 Ω measurement out of the range of operation. For this example, the next largest 1% resistor value above 1600 was selected.

Another benefit of setting the reference voltage to 1.62 V is that it sets the RTD measurement near the midpoint of the supply voltage. A reference of 1.62 V sets the input voltage for the ADC negative input. The input voltage is highest at the maximum RTD resistance is 0.4 V using an IDAC current of 1 mA and RTD resistance of 400 Ω . This sets the input voltage to 2.02 V for the ADC positive input.

Selecting a marginally larger resistance only reduces the resolution of the measurement. If the reference resistor is selected to be 2400 Ω then the reference voltage becomes 2.4 V. With an input to the ADC of 1.6 V (from 0.4 V after PGA gain of 4) compared to a reference voltage of 2.4 V, the ADC uses only 67% of the positive full-scale range.

1.5.4 Verify the Design Fits the Device Range of Operation

After determining the RTD range of operation, selecting the IDAC currents, the reference resistance, and the PGA gain, verify that the design still is within the range of operation of the device.

The PGA will have an input range dependent on the input common-mode voltage and the PGA gain. This may be different for each ADC. Determine the minimum and maximum input voltage and the common-mode voltage for each input voltage operation. By setting the input common-mode voltage to near mid-supply, the input voltage should be within the PGA range of operation. However, it is important to verify this through the equations given in the data sheet of the selected ADC. Consult the ADC data sheet for descriptions of the PGA and limitations in its input range.

Additionally, calculate the voltage at the output of the IDAC current sources. As the output voltage rises near the supply, the IDAC current will lose compliance as the output impedance of the current source is reduced. Calculate the voltage based on the IDAC currents driving the RTD resistance, reference resistance and bias resistance if necessary. If this voltage gets too close to the positive supply, the current may be reduced. Note that this compliance voltage will be different from device to device, and may vary by output current magnitude. Again, it is important to verify the compliance voltage based on the IDAC current source specifications in the data sheet of the selected ADC.

1.5.5 Design Iteration

If the design does not fall within the range of operation for the PGA, or is outside the compliance voltage of the IDAC, then another iteration the design may be necessary. It may be necessary to reduce or increase the reference resistance, or change any biasing resistors to set the PGA input range or set the IDAC output to with the compliance voltage.